

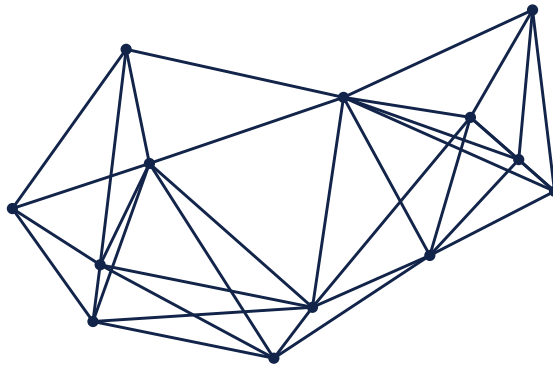
AI-POWERED CLINICAL DECISION SUPPORT • RESEARCH PROJECT BRIEF

Personalized Recommender System for Virus Research & Diagnosis Laboratory Network

Advancing Diagnostic Decision-Making through Artificial Intelligence

Amity Centre for Artificial Intelligence, Amity University, Noida, India

In partnership with ICMR – National Institute of Epidemiology, supported by the
Department of Health Research, Ministry of Health & Family Welfare, Government of India



PROJECT SUPPORTED BY THE INDIAN COUNCIL OF MEDICAL RESEARCH (ICMR)

JUMP TO

Approach & Methodology

Impact, Partnership & Team

PROJECT OVERVIEW

Turning Clinical Symptoms into Faster, Smarter Diagnostic Decisions

India's public-health laboratories confront a broad, constantly shifting spectrum of viral pathogens, where deciding which laboratory test to prioritise for a patient is a time-critical judgement call. This project — led by the Amity Centre for Artificial Intelligence with ICMR's National Institute of Epidemiology — turns a patient's symptoms and syndromic presentation into a ranked, confidence-scored shortlist of the most probable infections, guiding clinicians toward the right confirmatory test at the point of care.

- Smart Infection Triage**

Ranks the most probable infections and the right lab test from symptoms, demographics and syndrome.
- Rigorous Model Optimisation**

Multiple modelling approaches benchmarked for diagnostic accuracy and clinical consistency.
- Human-in-the-Loop Validation**

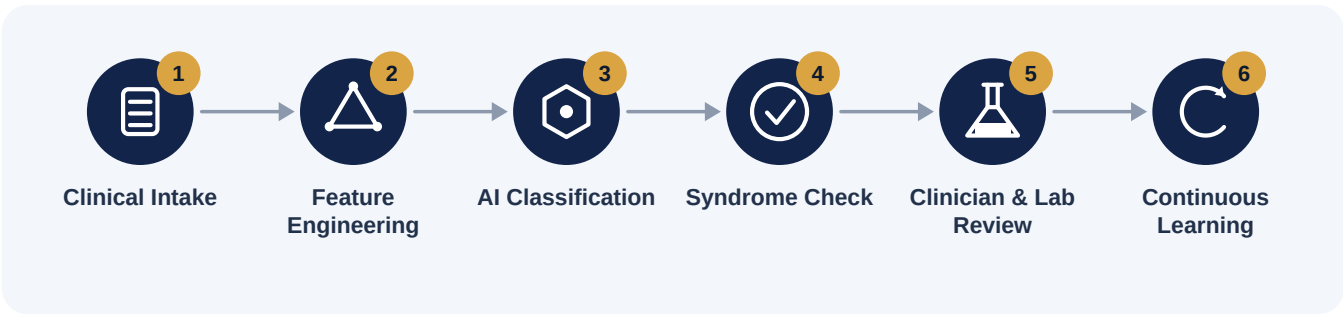
Clinician review and confirmed lab outcomes close the loop, strengthening the system with every case.
- Surveillance & Outbreak Intelligence**

Foundations for surfacing geographic and seasonal outbreak signals across ICMR's network.

METHODOLOGY

From Symptom to Recommendation: How the System Works

Each case moves through a structured, six-stage pipeline that pairs rich clinical data with a purpose-built AI model — and feeds confirmed outcomes back into the system.



Patient symptoms and geography feed a dual-stage AI classifier that narrows to the most likely infection group and sub-type, each with a calibrated confidence score. Recommendations are cross-checked against the reported clinical syndrome, reviewed by the treating physician, and confirmed against laboratory results — with every outcome feeding back into ongoing model evaluation.

DESIGNED TO SUPPORT, NOT REPLACE, CLINICAL JUDGEMENT

Every recommendation carries a calibrated confidence score and is reviewed by the treating physician against confirmed laboratory results before it informs patient care.

SCOPE & COVERAGE

Built on One of India's Richest Clinical Surveillance Datasets

Trained and continuously evaluated on real-world clinical data from ICMR's national surveillance network, spanning pathogens, geographies and syndromes.

24

Major infection categories

35

Structured clinical symptom indicators

9

Recognised syndromic categories

663K+

Curated clinical cases in the training corpus

100

Pathogens in the confirmatory reference panel

35+

States & union territories represented

INSTITUTIONAL PARTNERSHIP



Amity Centre for Artificial Intelligence

Amity Centre for Artificial Intelligence

ICMR NIE
INDIAN COUNCIL OF MEDICAL RESEARCH
NATIONAL INSTITUTE OF EPIDEMIOLOGY

ICMR – National Institute of Epidemiology

सत्यमेव जयते
Department of Health Research

Department of Health Research, Govt. of India

PROJECT TEAM

**Prof. M. K. Dutta**

Principal Investigator · Amity Centre for AI

**Dr. Rizwan S A**

Principal Investigator · ICMR, NIE

**R. Janani Surya**

Co-Principal Investigator · ICMR, NIE

**Dr. Rakesh C Joshi**

Co-Principal Investigator · Amity Centre for AI

**Dr. Abhishek Kaushal**

Project Scientist · Amity Centre for AI

**Shikhar Singh**

Contributor · Amity Centre for AI

PROJECT SUPPORTED BY THE INDIAN COUNCIL OF MEDICAL RESEARCH (ICMR) ·
amity.edu/noida/acai/research-projects.asp